The Story So Far

Random phenomenon
↓
Experiment
↓
$\Omega,\ \omega,\ A$
↓
$P(A)$
↓
Conditional probability / independence / counting
↓
$X : \Omega \to \mathbb{R}$
↓
Distribution of X : $\begin{cases} p_X(k), & \text{discrete} \\ f_X(x), & \text{continuous} \\ F_X(x), & \text{both} \end{cases}$
↓
Important numerical characteristics
↓
$E[X]$, $\mathrm{Var}(X)$, σ_X

$P(X \leq x)$

long-run average over repeated experiments.

$E[(X - E[X])^2]$ average squared deviation from its mean

$\sqrt{Var(X)}$

calculation:

$$Var[X] = E[X^2] - (E[X])^2$$

Properties of Variance

Suppose

$$Y = aX + b.$$

Then

$$E[Y] = aE[X] + b.$$

For the variance,

$$\operatorname{Var}[aX + b] = a^2 \operatorname{Var}[X].$$

Consequently,

$$\sigma_{aX+b} = |a|\sigma_X.$$

$\operatorname{Var}[Y] \overset{\text{defn.}}{=} E[(Y - E[Y])^2]$

$= E[(aX+b - E[aX+b])^2]$

$= E[(aX+b - aE[X] - b)^2]$

$= E[a^2(X - E[X])^2]$

$= a^2 E[(X - E[X])^2]$

$= a^2 \operatorname{Var}[X].$

Alternative way:

$\operatorname{Var}[Y] = E[Y^2] - (E[Y])^2$

Example [Sensor calibration and measurement error]

A computer-controlled system receives a measurement X from a sensor. Due to electrical noise and measurement uncertainty, X is a random variable. Suppose the sensor output has

$$E[X] = 3, \qquad \mathrm{Var}(X) = 9.$$

The system processes the sensor reading in two ways:

(a) It computes the squared error $(2 - X)^2$ relative to a reference value of 2. Find the **mean squared error** $E[(2 - X)^2]$.

(b) It applies a linear calibration $Y = 2X - 3$. Find the **variance of the calibrated signal** $\mathrm{Var}(Y)$.

(a) $E[(2-X)^2] = E[4 - 4X + X^2]$

$= 4 - 4E[X] + E[X^2]$

$= 4 - 4E[X] + Var[X] + (E[X])^2$

$= 4 - 4 \times 3 + 9 + 9$

$= 22 - 12$

$= 10$

(b) $Var[Y] = Var[2X - 3]$

$= 2^2 \cdot Var[X]$

$= 4 \times 9$

$= 36$

Moments of a Random Variable

The nth **moment** (or nth raw moment) of X is

$$E[X^n].$$

For example,

$$E[X]: \text{ first moment},$$
$$E[X^2]: \text{ second moment}.$$

The nth **central moment** is

$$E[(X - \mu_X)^n].$$

$\mu_X = E[X]$

Therefore,

$$\text{Var}[X] = E[(X - \mu_X)^2]$$

is the **second central moment** of X.

The terminology comes from mechanics, where moments describe how mass is distributed. Here, probability plays a role analogous to mass.

Where We Are Going

X
$\downarrow$
Distribution of X
$\downarrow$
$Y = g(X)$
$\downarrow$
Distribution of Y
$\downarrow$
Moments and other characterizations
$\downarrow$
Information carried by a random variable

Over the next two weeks, we're going to complete our study of a **single random variable** before moving on to multiple random variables.

Functions of a Random Variable

Question: If X is discrete (or continuous), must $Y = g(X)$ also be discrete (or continuous)?

Example 1 [Discrete to Discrete]

Suppose X represents the number of packet transmission failures, with possible values

$$X \in \{0, 1, 2, 3\}.$$

If the retransmission cost is

$$Y = 2X + 1,$$

then

$$Y \in \{1, 3, 5, 7\}.$$

Thus, Y is discrete.

In general, if X is a discrete RV taking values on $\{x_1, x_2, x_3, \cdots\cdots\}$, then $Y = g(X)$ must also be a discrete RV taking values on

$$\{g(x_1), g(x_2), g(x_3), \cdots\cdots\}.$$

Note some values $g(x_k)$ may repeat.

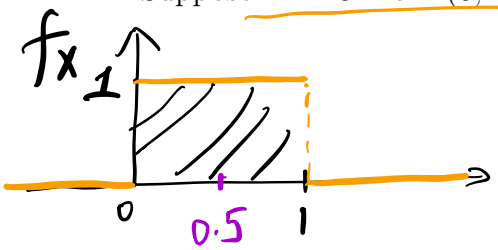
Then, the probability mass function of Y is

$$P_Y(y_i) = P(Y = y_i) = P(g(X) = y_i)$$

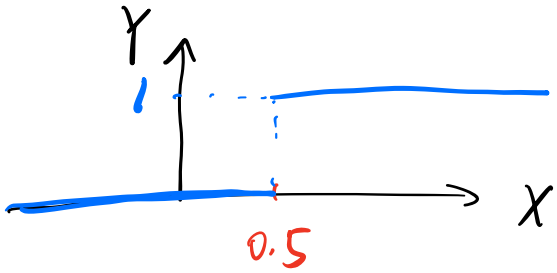
$$= \sum_{\forall g(x_j) = y_i} P_X(x_j)$$

Example 2 [Continuous to Discrete]

Suppose $X \sim \text{Uniform}(0,1)$, and a digital circuit converts an analog voltage into a binary output:



$$Y = g(X) = \begin{cases} 0, & X < 0.5, \\ 1, & X \geq 0.5. \end{cases}$$



$$P(Y=0) = P(X<0.5)$$

$$= \int_{-\infty}^{0.5} f_X(x)\,dx = \frac{1}{2}$$

$$P(Y=1) = P(X \geq 0.5)$$

$$= P(0.5 \leq X < \infty)$$

$$= \int_{0.5}^{\infty} f_X(x)\,dx$$

$$= \frac{1}{2}$$

Y is a discrete RV.

Note.

Suppose X is a continuous RV. If $Y = g(X)$ is a staircase function, then Y is a discret RV.

Example 3 [Continuous to Continuous]

Suppose

$$X \sim \text{Uniform}(0, 1)$$

and

$$Y = X^2.$$

Then Y is also cts. (check the example in Lec 12 for how to get its distribution).

Example 4 [Continuous to Mixed]

Suppose $X \sim \text{Uniform}(-1, 1)$, and

$$Y = \max\{X, 0\}.$$